SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15 13

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 50

Code of Conduct

I Siddhi Srivastava.P (192521454 (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Siddhi

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Total Marks:

CSA1513 - CLOUD COMPUTING FOR BIG DATA ANALYTICS : CO1 ASSESSMENT 2

CLOUD COMPUTING FUNDAMENTALS

DEFINITION

Cloud computing is the delivery of computing resources (like servers, storage, databases) over the internet, instead of owning and trying to manage them locally.



CHARACTERISTICS

- **On-demand self-service**
  - Users can access resources when needed.
- **Broad network access**
  Available over the internet, anywhere, anytime.
- **Resource pooling**
  Shared resources for multiple users (multi-tenant).
- **Elasticity**
  Scale up or down based on demand.
- **Measured service**
  Pay only for what you use (with monitoring)

BENEFITS

- **Cost Efficiency**
  - No hardware costs
  - Pay as you go.
- **Scalability**
  - Easily scale resources up or down.
- **Flexibility**
  - Access from anywhere
  - Supports multiple devices
- **Innovation**
  - Faster deployment
  - Access to latest technology.
- **High Availability & Reliability**
  - Redundancy and backup
  - Less downtime.

LIMITATIONS

- **Security and Privacy Risks**
  - Data breaches, loss of control
- **Dependence on Internet**
  - No internet = no access
- **Potential Higher costs**
  - Can increase with heavy usage.
- **Vendor Lock-in**
  - Hard to switch providers
- **Limited Control**
  - Less control over infrastructure.

BY:
SIDDHI

28)

EVOLUTION OF CLOUD COMPUTING

- From physical infrastructure to intelligent, on-demand services.

1. HARDWARE EVOLUTION



- Mainframes (1960s - 70s)
- Personal computers (1980s - 90s)
- Virtualization and x86 servers (2000s)
- Modern cloud data centers, edge devices, IOTs.

→ More powerful, smaller, energy efficient, scalable.

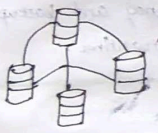
2. INTERNET SOFTWARE EVOLUTION



- Static web pages (1990s)
- Dynamic web and web applications (2000s).
- Web 2.0 (social, interactive)
- Mobile and cloud-based apps (2010s).

→ From simple browsing to connected, interactive services.

3. DISTRIBUTED SYSTEMS



- Single server → multiple servers.
- Load balancing.
- Fault tolerance
- Distributed databases.
- Global data centers.

→ Enables high availability, reliability and performance.

4. SERVICE DELIVERY



- IT as a service model
- On-demand self-service
- Pay-as-you-go.
- Elasticity and automation
- Global access.

→ From infrastructure ownership to flexible, utility-based services.

3)

SERVICE MODELS

Different levels of control, same goal - cloud convenience



IaaS

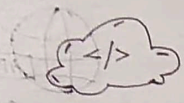
(Infrastructure as a service)

Provides virtualized computing resources. (servers, storage, networking)

Provider Responsibilities

- Physical data centers
- Virtualization
- Networking & storage
- Security & availability

Ex: AWS EC2
Microsoft Azure VMs
Google compute Engine



Paas

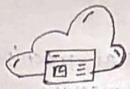
(Platform as a Service)

provides a complete runtime environment for building and deploying applications.

Provider Responsibilities

- Infrastructure
- Operating system
- Runtime & middleware
- Development tools.

Ex: Heroku
Google app Engine
Azure app service



SaaS

(Software as a Service)

Provides ready-to-use software applications via the internet.

Provider Responsibilities

- Application software
- Infrastructure and platform.
- Security & updates
- Availability and support.

Ex: Gmail
Microsoft 365
Salesforce.



Xaas

(Anything as a Service)

Broader model - any IT capability delivered as a service (Ex. DBaaS, Faas, STaaS) etc..

Provider responsibilities

- Core service infrastructure.
- Management and scaling.
- SLAs and support.
- Security, monitoring

Ex: DBaas (AWS RDS)
Faas (AWS lambda)
STaas (AWS storage)

A9)

Virtualization and Web Services as Enablers of cloud

Abstract resources + Standardized communication = Flexible, Scalable cloud

Virtualization

Creates multiple virtual machines on a single physical machine.

Key features:

- Resource abstraction
- Isolation & security
- Better utilization
- Live migration

→ Allows flexible, on-demand allocation of resources.

Web Services

Enables communication between applications using standard protocols.

Key features:

- Standard protocols (HTTP/HTTPS)
- XML / JSON data format
- Interoperability
- Service discovery.

Cloud Computing
(On-demand - Elastic - Scalable)

Enables cloud deployment & scalability

Elastic Scaling	Interoperability & integration	High availability & reliability	Cost efficiency.

9/10/26